

Rock Jobs: Which Property Matters?

I can identify which tested or observed rock property matters for a stated everyday job.

JOB

what must the material do?

PROPERTY

what can we observe or test?

WHY

connect the evidence to purpose

[Open the original interactive lesson](#)

What success looks like

- State the job before looking at candidate samples.
- Choose a relevant property such as scratch resistance, water passage, shape or appearance.
- Explain why one property helps the job and why no single property decides every use.

Retrieve and reconnect

- Last lesson: what did the water-passage result measure?
- Last lesson: why did a lower value not mean waterproof forever?
- Last week: what did the scratch comparison tell us?
- Lead-in: name one property that could matter for a walking surface.

A pupil says, 'The least permeable sample must be best for every rock job.' Which first idea would you test?

Think first. Point, say, sign or write your choice. Give one reason.

A. Agree. Every use needs the least water passage.

B. Ask what the job is; a rain barrier and a drainage layer need different water behaviour.

C. Ignore water and choose the brightest colour for every job.

Look closely · what is the evidence?

Roof-cover brief: help stop rain reaching the space below.

Hard-wearing path brief: cope with repeated contact and provide a suitable surface.

Drainage-layer brief: allow water to move away through spaces between suitable rock pieces.

Look closely · what is the evidence?



property that matters: water passage

A roof job card



property that matters: scratch resistance

A path job card



property that matters: gaps between pieces

A drainage job card

Words we will use

- property — what the material is like
- purpose — the job it must do
- suitable — works for this job

Watch the model

- Read the brief before the samples. A rain-facing cover, hard-wearing path, indoor display and drainage layer do different jobs.
- Turn each job into a property question: How much water passes? How well does the surface resist scratching? Can pieces form the needed shape? Does appearance matter here?
- Match only relevant evidence. Colour does not answer water passage, and a permeability result does not prove resistance to wear.

Which property question is most relevant to a rain-facing roof-cover model?

- A. Which sample is the darkest?
- B. Which prepared sample lets less water through and can form overlapping flat pieces?
- C. Which sample has the longest name?

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- A job can need more than one property. A path may need resistance to surface wear and a suitable walking surface; a display may value pattern while still needing stability.
- The direction of useful can change. Low water passage may help a rain barrier, while spaces between suitable pieces can help a drainage layer move water away.
- If the key property has not been observed or tested, mark needs another check. A confident guess is not a material decision.

Find and repair the misconception

Rock-job glitch: the least permeable or hardest sample is best for every purpose.

A. Keep it: one winning property settles every job.

B. Repair it: start with the purpose and choose the property evidence that answers that job.

C. Repair it: replace tests with colour preference.

Job-to-property question lab

Route each brief clue to WATER CONTROL, WEAR / SURFACE, SHAPE / APPEARANCE or NEEDS ANOTHER CHECK.

- Read the job clue.
- Ask what must happen or be prevented.
- Choose the relevant property question.
- Freeze two links and explain why they fit.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- WATER CONTROL
- WEAR / SURFACE
- SHAPE / APPEARANCE
- NEEDS ANOTHER CHECK

Say: The job needs _____. The property question is _____ because _____. We still need to check _____.

Your independent mission

Create property-question cards for four everyday rock jobs before any candidate sample is chosen.

Record: Four job-property questions, two because explanations, one needs-another-check decision and one model limit.

Choose your support and challenge

- Supported: Match one key property picture to three clear jobs and complete one because sentence.
- Standard: Complete all four briefs, naming the relevant property evidence and one missing check for each.
- Stretch: Explain a two-property trade-off and why opposite permeability behaviour can suit different water jobs.

Show what you know

Explain which property matters for one job and why another property does not answer it.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.